Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Taylor's and Maclaurin's expansion

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Taylor's and Maclaurin's expansion

1 Taylor's and Maclaurin's expansion

Study focus: Taylor's and Maclaurin's expansion. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today you will be learning an important concept called the Taylor's and McLaren series, which are very helpful in reducing any complicated problems in the form of a polynomial, and it has a lot of applications in engineering.

Key details

- But if I don't let you use a calculator, you think that she has gone crazy and then, you don't know what's tan of cube root of three, and then on that I'm asking you to find exponential power of that.
- What goes behind that?
- Anything that is complicated.
- It will not understand your differentiation of sine, cos, or, if you want to give a function that's like e to the power of tan of x square and then ask you ask the calculator to differentiate it.
- What it does every time is it try to express any complex function as a series of polynomials.
- Few of them, I don't know, you might be aware because I think you are aware of binomial expansion.
- But if I make a small change in this, I'll give sine x square dx.
- Then what is the integral of this?
- Maybe if you want, I'll give you definite integral.
- Try plugging it this in the calculator because anyway you cannot find the value of this directly.
- Then the calculator writes this sine of x square in the form of a polynomial up to few terms, depending on the configuration of the calculator.
- And then it calculates once you get a polynomial, everything is easy, right?

- If I can write this as a polynomial, you integrate every term, apply the limits, simplify, you will get the answer.
- That is what is happening.
- Or as simple as I have mentioned here, e to the power of x square, which you don't know how to integrate.
- Zero by zero, and then you use L'opitholes rule, you did learn mean value theorems like Cauchy's and Legrange and Rolls.
- And you might be wondering how did they thought about those theorems?
- And m even though we are not discussing Taylor's theorem, but Taylor's series is nothing but consequence of Taylor's theorem.
- Why am I adding this continuous?
- Why am I prefixing this derivative with continuous?
- That's because I want to have a derivative of n plus one th order.
- I'm not just finding the first derivative, the second derivative, or the third derivative.
- In fact, I'm finding the n plus one order derivative.
- Then Taylor says that you can expand this function around any point A using a polynomial.
- So that means it looks like f of x, I can write this as f of a plus x minus a into f dash of a.

Example or application from the lecture

- suppose I'll ask you to,, find the value of this function, e to the power of tan of cube root of three.
- Another thing is suppose I'll ask you to evaluate this integral.
- Let's say I'll give you integral of sine of x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

Plus x minus a the whole square divided by two factorial into f double dash of a and then this continues.

Key details

- This is not new to you, right?
- This is an infinite series, of course, but don't see that.
- Aren't you familiar with this?
- What is this?
- I can write rewrite this expression as f of x minus f of a divided by x minus a, which is nothing but f dash of a, right?
- Instead of expanding it around a point A, if I expand the function about the origin, about the origin, then we get the McLaren series.
- And once I have a polynomial, I can do everything with it.
- I can differentiate, I can study what is the change that is happening, I can integrate, I can find the limit of that.
- Whatever you ask me to do, I can do it.

- That's why these please remember that Taylor's in McLaren series always comes very handy for you when you have to handle complicated conditions or complicated functions.
- Mechlerin series, which you might already know, right?
- I to add a few, you can write.
- The series for this as well, which is 1 by 1 minus x.
- And one surprising thing is if they give me this series, I know that I'm getting a finite value.
- And that's very strange.
- Actually Ramanujan proved so many such results.
- Because you are adding infinitely many positive terms and it's coming back to minus one, which is not that easy to accept.
- what is happening there, right?
- But I want you to explore some of those things.
- I have log of 1 plus x, you can try log of 1 minus x, sine tan, hyperbolic, tan inverse.
- But how the question is, how are they getting this?
- It's very simple.
- I'll anyway, I won't be solving problems on Taylor's and McLaren series for one variable because I'll be extending it to bivariate function.
- But I can show you what happens.
- And I know that derivative of e par x is e par x itself.
- No matter how many times I differentiate, I'll get e power x only.
- And I can substitute these, replace this all these x's with 0.
- And that's not new to us, right?

Example or application from the lecture

- Suppose I'll show you why it is not new.
- Suppose I'll erase all the higher degree terms, okay?
- That means given any function, I can write this as a polynomial of infinite order with infinitely many terms.

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Core explanation · 10:00–14:59

All you need to do is find the value of the function at 0, provided f of 0 does not tend to infinity.

Key details

- I cannot write equal to because we cannot reach infinity.
- That's not defined because every at all the derivative at zero will tend to infinity.
- That's the only constraint.
- And here the same thing f of a cannot be infinite.
- Cannot tend to infinity or cannot be equal to infinite.
- And then I want to express it in terms of a polynomial.
- If you want about a point x is equal to a, then you use Taylor series, and if you want about the origin, you use metogen series.

- That's all you want, right?
- As I said before, it you can use this to evaluate any complicated function by expressing it in terms of a polynomial.
- You can evaluate definite integrals, which I also discussed, right?
- 0 to 1 sine of x squared.
- And if you want sine x square, just replace x with x square.
- That's all you need to do.
- Minus x to the power of 6 by 3 factorial plus x to the power of 10 by 5 factorial minus etc.
- And the question says you can you have to integrate this from 0 to 1.
- So that's easy now for me because I know what is integral of x square, x par 6, x par 10, etc.
- I can integrate all these things with respect to x and then apply the limits.
- That's all I need to do.
- I'll show you that when we plot sign any function and its series, okay?
- The next thing is understanding asymptottic behavior of the functions.
- that these are the asymptotes for the function, right?
- In this course, you will be learning higher order differential equations and when you need to solve the higher order differential equation, we will be basically using the Taylor series.
- And in physics, McLaren series used to approximate the Lorentz factor.
- And E is equal to M C square.
- The most famous equation E is equal to M C square.
- It's an approximation on velocities, which is done again by the McLaren series.
- In the motion of a pendulum, we approximate sine theta to be theta, that itself is a that means in the series expansion of sine theta.
- You're taking only one term.
- And again, it's used in power flow analysis of electrical power systems and can be used to find generating functions.
- This is a very, very important concept for computer science students.
- Okay, but anyway, when we do four-year series in the next semester, we'll talk about generating functions what it does and what do we mean by generating functions and other things.

Example or application from the lecture

- That means if of suppose I want to find the McLaren series for 1 by X.
- Suppose I have a rational function here and I want to check what is happening to its asymptotes.

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Core explanation · 15:00–19:59

And the graph that is in red is the McLaren series for that.

Key details

- What is the series for sine x?

- Sine x is x by 1 factorial minus x cubed by 3 factorial and then plus x power 5 by 5 factorial and then etc.
- This blue graph is the graph of sine x.
- And then you see I have taken only four terms here.
- I have taken the maybe one more I'll add.
- in fact this is let me change this to blue so that I'm plotting sin x in blue.
- The next graph that you see.
- is again n is equal to 6.
- And see, this is maybe this is n is equal to 10.
- I'm just adding 10 terms in that polynomial, which is not too many as compared to an infinite series, right?
- It's overlapping.
- That's why it works so best.
- The first advantage is I'm getting everything in terms of a polynomial.
- That's the first advantage.
- I'm fine with that, I'm happy.
- The second thing is once it you are telling me to take infinite series, but most of the series converges for finitely many terms.
- Maybe not as m small as sinex, you might have to take say around 40 or 50 or whatever the number might be.
- That depends on the function.
- Again with the condition that this f of a comma b cannot go to infinity.
- plus this is f dash plus x minus a the whole square by two factorial into f double dash of n etc.
- I didn't talk much about this because you are already familiar with the binomial theorem, right?
- you already know what is binomial theorem expansion is and then how do you get the coefficients and other things.
- So for a Taylor series for a function of two independent variables will look like this f of ab plus x minus a into f x of a b because there are two derivatives.
- You have to take x minus a for fx, y minus b for f y.
- And there are four second order partial derivatives, fxx, fxy, and then f y.
- But that they are equal.
- F x y is the same as f yx.
- That's why you got this two.
- And then of course the next term will continue.
- The only thing I need to worry is not just this, but any of these cannot tend to infinity.
- I did mention only this but not necessarily.
- None of the derivatives fx, f y and f x, f y.
- At zero, zero, of course.
- At zero, zero.
- And it's easy to remember because fx goes with x.
- f y goes with y, f x goes with x square, fx y goes with 2xy I'm getting this too because mixed second order partial derivatives are equal and then f y goes with y square.

Example or application from the lecture

- But definitely the function matches exactly with the given function in terms of tailors or in terms of material series.
- Instead of AB I have to consider the origin.

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Core explanation · 20:00–24:59

So,, before we solve some problems, this is the graph of polynomial because it's two dimension two variable I have to go to three d so it's not as good as visualizing in just one dimension.

Key details

- But anyway I have done my best.
- I have written a Taylor polynomial of this.
- so of just the first degree.
- and that's what I get.
- Maybe they have used the McLaren series.
- And now I'm taking the second degree polynomial.
- And then we'll see how many terms we need to take so that they're almost together.
- These are some of the applications of Taylor series for two variables.
- Basically, what they do is they just take a they are going to make use of the linear and optimization process.
- and distances can be measured accurately, variance of complicated configuration and as I said, visualization of complicated functions and so geometrically it's just that.
- In the next session, we'll try to solve some problems on Taylor's and McLaren series for a function of two independent variables.

Example or application from the lecture

- We can express any given function in terms of its polynomial.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Taylor's and Maclaurin's expansion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00044	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

C Revision and retrieval practice